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COURSE PROJECT GUIDELINES
IT 223 – Integrative Programming and Technologies 1

Basic System with API Integration

Project Overview:

For this project, you will develop a basic software system that integrates data from a public API. The goal is to demonstrate your understanding of how applications communicate with external services.

You are free to choose any programming language, framework, or platform, as long as your system successfully consumes and displays data from an API.

Learning Objective:

By completing this project, you should be able to consume data from a public API, process and display the API responses within your application, handle basic errors such as invalid requests or lack of internet connection, and build a simple but functional system that demonstrates proper API integration.

Project Requirements:

1. System Requirements

Your system must:

- Use at least **one public API**
- Send a request to the API
- Receive and process the response
- Display the retrieved data in a user-friendly way
- Include basic error handling
- Have a simple user interface (GUI or web-based)

2. Allowed Technologies

You may use any of the following (or similar):

Programming Languages

- Python
- Java
- JavaScript
- C#
- PHP
- Dart (Flutter)
- Other languages

Optional Tools

- cURL
- requests (Python)
- fetch/axios (JavaScript)
- Postman (for testing)

3. **Suggested Project Ideas**

You may create your own idea, or choose from these:

- Weather Viewer
- Movie Search App
- News Reader
- Random User Generator
- Currency Converter
- Cat/Dog Image Viewer
- Book Search System

Note: Creativity is encouraged.

Minimum Features:

Your project must include:

Functional Features

- API request button or automatic fetch
- Display of API data
- Search or input field (if applicable)
- Proper error messages

Technical Features

- Clean and readable code
- Proper variable naming
- Comments in important parts
- Organized project structure

Required Deliverables:

Submit the following:

1. **Source Code**
   - Complete and runnable
   - Properly organized

2. **Project Documentation (PDF or DOCX)**
   Include:
   - Project title
   - Student name(s)
   - System description
   - API used
   - Screenshots of the system
   - How to run the program

3. **Demo (3–5 minutes)**
   Show:
   - System overview
   - API working
   - Key features

Grading Rubric:

Criteria	Percentage
Functionality	30%
API Integration	25%
User Interface	15%
Code Quality	15%
Documentation	10%
Demo	5%
TOTAL	**100%**

Important Reminders

- Do **not** hardcode fake API data
- Respect API rate limits
- Use free/public APIs only
- Test your system before submission
- Plagiarism is strictly prohibited

---------- *NOTHING FOLLOWS* ----------